GenAI Introduction Report

AI & Data Training

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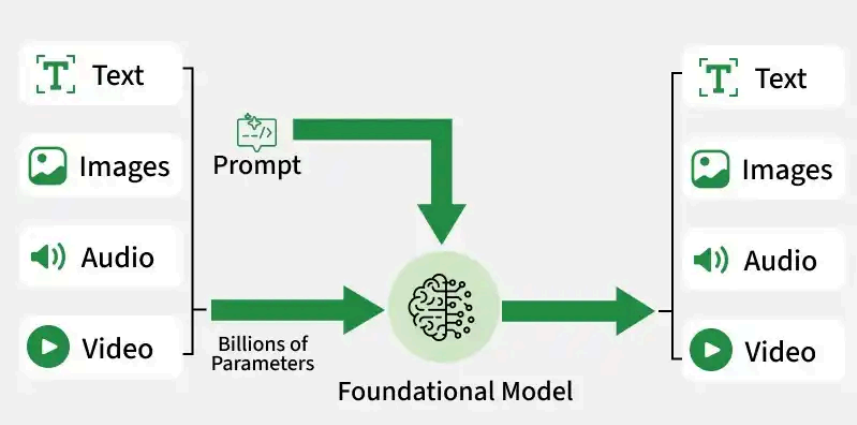
Outline

- *What is Generative AI?*
- *How Generative AI Works?*
- *Types of Generative AI Models*
- *Types of Generative AI based on Output and their Use cases*

What is Generative AI?

Generative AI is a type of artificial intelligence designed to create new content such as text, images, music or even code by learning patterns from existing data. These models generate original outputs that are often indistinguishable from human-created content. These models use techniques like deep learning and neural networks to generate output.

Unlike discriminative AI which focuses on classifying data into categories like spam vs. not spam, generative AI creates new data such as text, images, audio or video that resembles real-world examples.



How Generative AI Works

1. Core Mechanism (Training & Inference)

Generative AI is trained on large datasets like text, images, audio or video using deep learning networks. During training, the model learns parameters (millions or billions of them) that help them predict or generate content. Here models generate output based on learned patterns and prompts provided

2. By Media Type

- Text: Uses large language models (LLMs) to predict the next token in a sequence, enabling coherent paragraph or essay generation.
- Images: Diffusion models like DALL·E or Stable Diffusion start with noise and iteratively denoise to create realistic visuals
- Speech: Text-to-speech models synthesize human-like voice by modeling acoustic features based on prompt.
- Video: Multimodal systems like Sora by OpenAI or Runway generate short, temporally coherent video clips from text or other prompts.

3. Agents in Generative AI

Modern systems often use agents which are semi-autonomous components that interact with the environment, obtain information and execute chains of tasks. These agents use LLMs to reason, plan and act enabling workflows like querying databases, performing retrieval or controlling external APIs.

4. Training and Fine-Tuning

LLMs are trained on massive general corpora (e.g., web text) using self-supervised methods. These models become pre-trained models which can be further trained on domain-specific labeled data to adapt to specialized tasks or stylistic needs. This technique is called fine tuning, and it can be done using:

LoRA, QLoRA, Peft, Reinforcement Learning from Human Feedback (RLHF), LLM Distillation

5. Retrieval-Augmented Generation (RAG)

Modern systems also use RAG which enhances outputs by retrieving relevant documents at query time to ground the generation in accurate, up-to-date information, reducing hallucinations and improving factuality. The process typically involves:

- Indexing documents into embeddings stored in vector databases
- Retrieval of relevant passages
- Augmentation of the prompt with retrieved content
- Generation of grounded, informed responses

This approach preserves the base model while enabling dynamic knowledge updates.



Types of Generative AI Models

1. Transformers or Autoregressive Models

- Autoregressive Transformers Models generate sequences by predicting the next token based on all previous ones moving step by step through the text.
- The architecture relies on the transformer's self-attention mechanism to capture context from the entire input so far making it highly effective for natural language and code

generation.

- Popular examples include GPT models which can produce coherent, context aware paragraphs, solve coding tasks or answer complex queries.
- The autoregressive approach gives fine grained control over each output step but can be slower for long generations since tokens are generated one at a time.

2. Diffusion Models

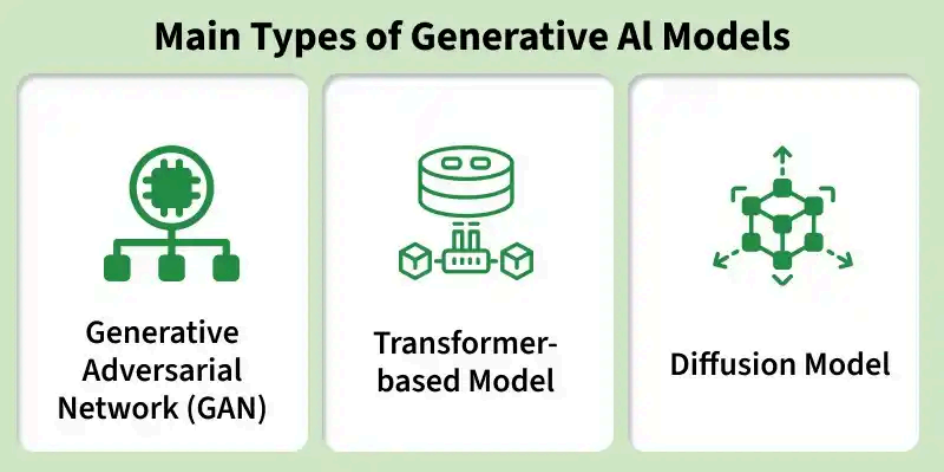
- Diffusion models generate data such as images or audio by starting with pure random noise and gradually refining it into a coherent output through a series of denoising steps.
- Each step reverses a simulated diffusion process that added noise to real data during training.
- This iterative approach can produce highly detailed and realistic results specially in image synthesis where models like Stable Diffusion and DALL·E 3 have set benchmarks.
- Diffusion models are also versatile they can be adapted for inpainting, style transfer and conditional generation from text prompts.

3. Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs)

- VAEs and GANs were among the first deep learning architectures for generative tasks.
- A VAE encodes data into a compressed latent space and then decodes it back with a probabilistic twist that encourages smooth, continuous representations. This makes them good for controllable generation and interpolation between styles.
- GANs in contrast use two networks against each other a generator that tries to produce realistic outputs and a discriminator that tries to detect fakes.
- This adversarial setup leads to sharp, lifelike images though training can be unstable and prone to mode collapse.

4. Encoder Decoder Models

- Encoder decoder architectures consist of two stages: the encoder processes the input into a dense representation, and the decoder generates the desired output from that representation.
- They are widely used for sequence-to-sequence tasks like language translation, summarization and image captioning.
- The encoder captures the full context of the input before the decoder starts producing tokens, allowing for strong performance on tasks that require global understanding rather than token by token prediction.
- Modern encoder decoder models often use transformers for both stages as in T5, BART and many multimodal systems.



Types of Generative AI Based on Output and Their Use Cases

Generative AI models are categorized based on the type of output they produce and their use cases. Here's a breakdown of the types and their applications:

- **Text Generative Models**: These models generate text content, such as articles, stories, or code. They are widely used in content creation, writing assistance, and programming.



- **Image Generative Models**: These models create images, including photos, illustrations, and graphics. They are used in content creation, design, and visual storytelling.



- **Audio Generative Models**: These models produce audio content, such as music, sound effects, or voiceovers. They are utilized in music production, audio editing, and virtual reality.

- **Video Generative Models**: These models generate video content, including animations, tutorials, and commercials. They are applied in video production, training, and entertainment.

- **Code Generative Models**: These models create code, including programming languages, scripts, and software. They are used in software development, programming assistance, and game development.

Each type of generative AI model has its own set of use cases and applications, making it essential to choose the right model for specific tasks and industries.